The Relationship Between City-Level Gentrification and Crime Rates: A Study of French Cities (2016-2021)

Léo-Paul Garelli and Noé Mansuy

GitHub Repository: https://github.com/LEOPAUL55/Noe_Leo_final

Research question

Gentrification, theorized in the 1960s, is an urban process in which higher-income groups move into and transform historically working-class neighborhoods. This shift is typically marked by rising real estate prices and an increase in the city or the neighborhood's average standard of living. As wealthier residents renovate properties and demand grows, housing costs rise, often displacing lower-income residents to less central areas or other cities. This leads to social conflict that affects education, inequality, health, and crime.

This project examines whether gentrification in French cities correlates with a decrease in recorded crime within those municipalities, and whether crime is displaced to non-gentrified areas. Our goal is to assess the short- and medium-term impact of gentrification on crime across France, to inform more effective public policies.

We analyze public data from 897 French cities between 2016 and 2021. First, we construct a gentrification index to measure the intensity and geographic spread of the phenomenon across the period, allowing us to classify cities by their level of gentrification. Second, we use this index to estimate the direct and delayed effects of gentrification on crime trends—both overall and by crime type—in each municipality.

Datasets descriptions

To ensure reliable findings, we begin by building two core datasets. The first contains annual data by city (e.g., real estate prices, median income, poverty rate, and crime counts). The second, the evolution dataset, is derived from the first and will track changes over time.

The gentrification index is constructed using economic indicators per city per year: real estate price per square meter, median income, the poverty rate (below 60% of median income), and the first and ninth income deciles. For crime, we use the annual number of recorded crimes by type. Using population data, we then calculate overall and type-specific crime rates.

Data sources

1. **Real estate price - Datagouv / Chambre des notaires de Paris**

The informations about the real estate price per city are available on the France state's site. They are given in the files called "Demande de valeur foncière" for every year, that is from 2016 to 2021. This is a collection of various variables on residential housing market at the city level between 2014 and 2024. What is interesting for us is the mean price by square meter of the goods that have been sold each year. This was published by Boris Mericksay. Unfortunately, the informations about Paris at the arrondissement level are missing in those files. These important informations can be found on the Chambre des notaires de Paris' site that gives the trimestrial average square meter price of selling. For simplicity, we only focus on the selling price of appartements in the city from 2016 to 2021 (trimester 4). The available file is a PDF that can be easily convert into a CSV file.

2. **Average standard of living - Insee**

The files "poverty" are collections of indicators of disposable income in France, at the city level, between 2016 and 2021. We will only focus on the poverty rate, the median standard of living, the first and the ninth deciles. Poverty data are provided as separate yearly files with heterogeneous structures. Variable names differ across years, and the 2018 data are reported at the IRIS level rather than the commune level. To handle this, we implement a loading strategy that reads each file separately while restricting imports to the variables of interest (years 2016:2021). Poverty indicators are harmonized across years. Non-numeric values are explicitly converted to missing values during loading, but this will not affect our analysis because this only concerns small cities. The result is a consistent panel of indicators comparable across years.

3. **Criminality - Datagouv / Ministère de l'intérieur**

The CSV file "criminality" is a database of all the crimes and offences (except traffic offences) at the city level in France, published by French Police Force, since 2016. These offences were identified following a complaint filed by a victim, a report, a testimony, a flagrant offence, a denunciation, but also on the initiative of the security forces. The total number of crimes per year per city is computed by doing the sum of each type of crimes for every year. Thanks to the informations about the number of inhabitants per city given in the past files, it is is possible to compute the global crime rate and per type.

Table 1: Variables in the final dataset

Variable	Type	Description
CODGEO	character	Unique city code (first 2 digits = department)
MED	numeric	Median income
TP60	numeric	Poverty rate (< 60% of median income)
D1	numeric	1st income decile
D9	numeric	9th income decile
year	numeric	Year
IRIS	character	City population
insee_pop	integer	Domestic physical violence
Violences physiques intrafamiliales	integer	Non-domestic physical violence
Violences physiques hors cadre familial	integer	Armed robberies
Vols avec armes	integer	Violent robberies (no weapon)
Vols violents sans arme	integer	Non-violent robberies
Vols sans violence contre des personnes	integer	Home invasions
Cambriolages de logement	integer	Vehicle thefts
Vols de véhicule	integer	Thefts from vehicles
Vols dans les véhicules	integer	Thefts of vehicle accessories
Vols d'accessoires sur véhicules	integer	Vandalism
Destructions et dégradations volontaires	integer	Drug use
Usage de stupéfiants	integer	Drug dealing
Trafic de stupéfiants	integer	Fraud
Escroqueries et fraudes aux moyens de paiement	integer	Total crimes
total_crimes	numeric	Price per m^2
Prixm2Moyen	numeric	City name
LIBGEO	character	Crime rate per 100 inhabitants
crime_rate	numeric	Unique city code (first 2 digits = department)

Descriptive analysis of the crime rate evolution from 2016 to 2021 - comparison between violent and non violent crimes

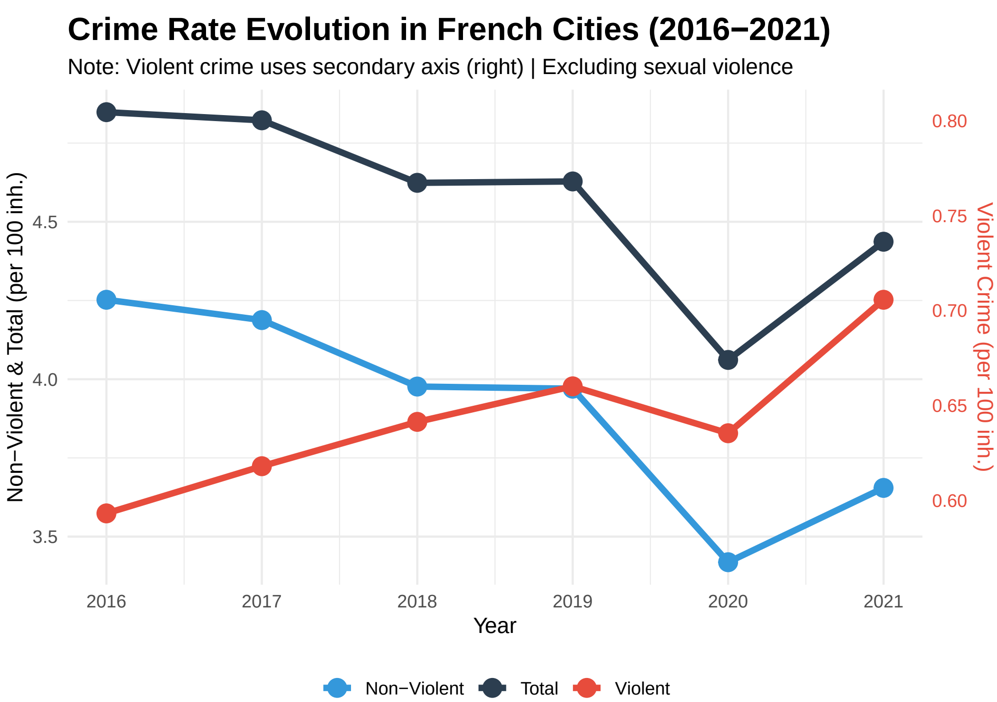
To analyze whether gentrification impacts violent and property crimes differently, the original 13 crime categories are grouped into two broader types:

Violent crimes: domestic and non-domestic physical violence, armed robbery, and unarmed violent robbery.

Non-violent crimes: theft, vehicle-related crime, burglary, vandalism, fraud, and drug-related offenses.

Between 2016 and 2021, total crime and non-violent crime follow a similar downward trajectory, while violent crime increased.

Note on exclusions: Sexual crimes and drug use offenses (usage de stupéfiants, AFD) were excluded from this analysis and from the total crime figures. Sexual crimes were removed because reporting trends are heavily influenced by social and cultural factors (e.g., the #MeToo movement), making them unreliable for standard temporal comparison. Drug use offenses were excluded as they were not reported at all during the first years of the period studied



Building a gentrification index between first year and last year :

After obtaining clean data sets, we can now analyse it through several aspects. The first step is to build a gentrification index, that is a mathematical formula which enables to compute and compare the degree of gentrification per cities thanks to the available information in the data set (economic description, real estate price). Second, it will be possible to compute descriptive statistics about gentrification to have a better understanding of the process in France through the period. More precisely, to what extent are french cities affected by gentrification and where is the process the most intense on the map. In the same idea, descriptive statistics about crime in France will be computed to also understand this phenomenon and its evolution. Third, we will try to isolate with the data the causal link between the degree of gentrification and the criminality evolution in the cities using an econometric method.

Recalling the definition, gentrification is the process by which working class quarter are appropriated by upper classes, which exclude the initial population and replace it by a wealthier one. In the data, such a process can be observed using the data sets “ppsm” (real estate price) and “poverty” (economic information). Indeed, a gentrified city can be identified through a rise in the real estate price (Prix2Moyen) which will increase the city’s standard living and rise the median income (MED) as well as the first and last decile (D1 and D9) and reduce the rate of poor people in the population (fall of TP60).

Using these data, we can create a gentrification index that is implementable to any city using the following formula:

Gentrification = (**price_var** * 0.25) + (**MED_var** * 0.20) + (**D9_var** * 0.10) + (**D1_var** * 0.35) + (**TP60_var** * 0.10)

Gentrification is not about the level of the variables, but their variation. Thus we need to compute the variation in percentage of each variable to implement the formula. Moreover, we justify the weight given to each variable thanks to the literature. According to the initial definition (Ruth Glass, 1964) combined with the work of Neil Smith and the Rent-Gap Theory (1979), gentrification mainly materialized by the displacement of the working class as well as the rise in the real estate price, that justifies the heavy weight on the first decile (D1_var) and the square meter price (price_var). To make the index meaningful, we standardize it using the z-score standardization and center it around 0 (if the index is equal to 0, the city is as gentrified as the national mean). Finally, we exclude the city that have a median income greater than 60% of the national one, considering they are already composed of upper classes, so it cannot be gentrified.

Table 2: Top 10 Most Gentrified Cities (2016-2021)

City Code	City Name	Gentrification Index
91021	Arpajon	1.867
92024	Clichy	1.726
93048	Montreuil	1.526
75119	Paris 19e Arrondissement	1.499
59163	Croix	1.487
93006	Bagnolet	1.394
59410	Mons-en-Baroeul	1.345
54547	Vandoeuvre-lès-Nancy	1.313
93001	Aubervilliers	1.287
61169	Flers	1.274

Cross sectional analysis

After constructing the gentrification index for all eligible French cities (those with baseline median income below the 60th percentile of the national distribution), we proceed to examine the relationship between gentrification intensity and changes in crime rates over the 2016–2021 period. The analysis is cross-sectional: for each city, we compute the percentage change in crime rate between the first and last year of observation, and correlate this change with the standardized gentrification index.

To facilitate interpretation, we categorize cities into three groups based on their gentrification index:

Low gentrification: bottom 25% of the index distribution

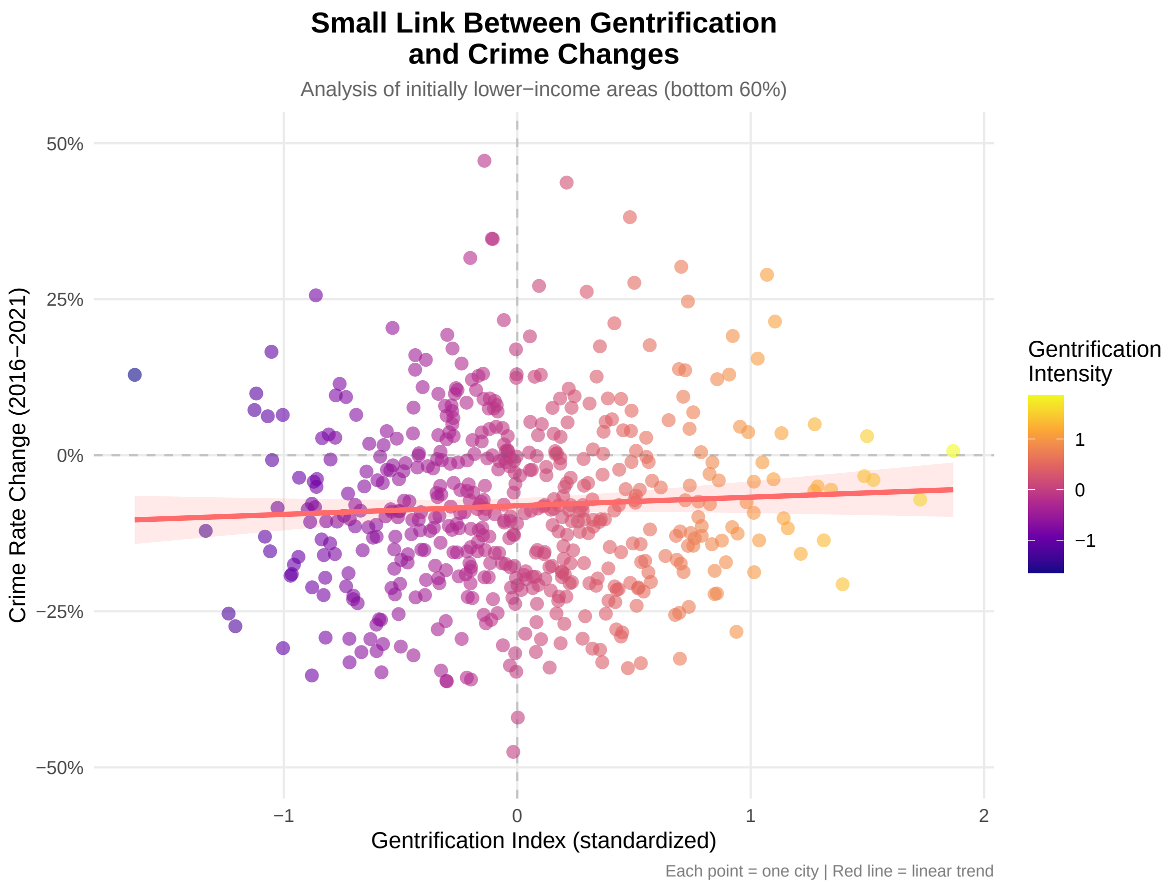
Medium gentrification: middle 50%

High gentrification: top 25%

These categories allow us to compare how crime evolved differently across areas experiencing varying degrees of socioeconomic transformation.

Visual exploration: Scatter plot

The scatter plot below displays the relationship between the gentrification index (horizontal axis) and the percentage change in crime rate (vertical axis) for all 897 cities in our sample.



Visually, a slight positive trend emerges: cities with higher gentrification indices tend to experience smaller reductions (or even increases) in crime rates compared to less gentrified cities. The cloud of points is dispersed, suggesting considerable heterogeneity. To formally test whether this visual pattern reflects a true association, we use a Pearson correlation test—the standard method for assessing linear relationships between two continuous variables. The results show no significant linear relationship between gentrification intensity and crime rate change ($r = 0.052$, $df = 535$, $p = 0.227$).

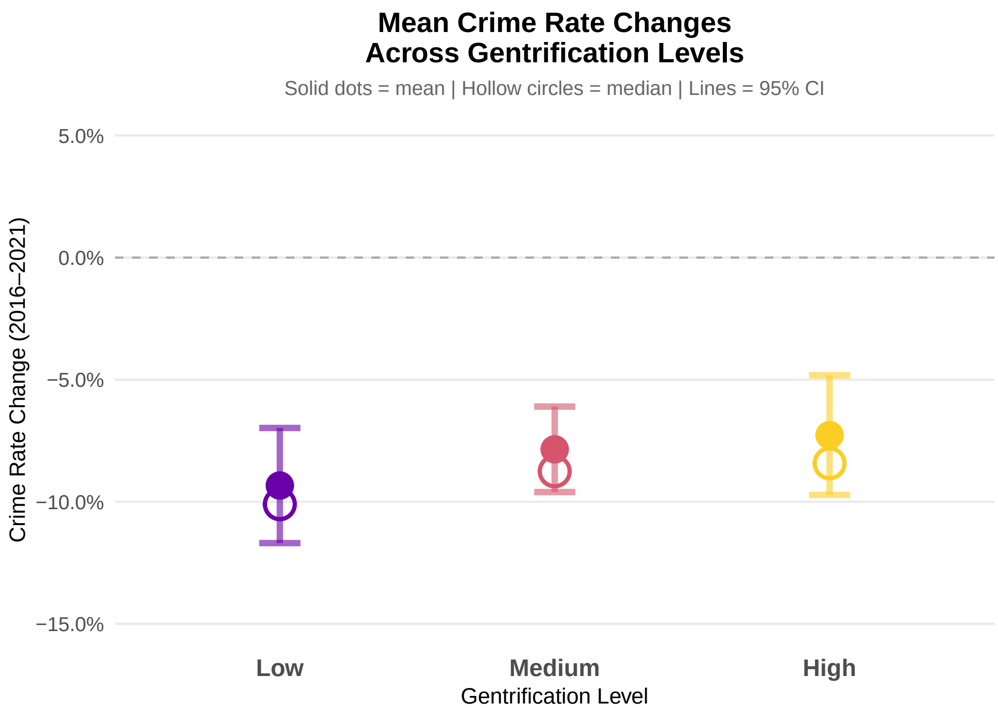
Table 3: Pearson Correlation: Gentrification Index vs. Crime Rate Change

Statistic	Value
Pearson correlation (r)	0.0522
t-statistic	1.2100
Degrees of freedom	535.0000
P-value	0.2270
95% CI (lower)	-0.0325

Statistic	Value
95% CI (upper)	0.1362

Comparing crime rate changes across gentrification levels

To better quantify these differences, we compute the mean and median crime rate changes for each gentrification category, along with 95% confidence intervals.



All three categories experienced a reduction in crime rates between 2016 and 2021, consistent with national crime trends in France during this period. While the magnitude of reduction appears to vary visually across gentrification levels, we need to formally test whether these differences are statistically significant.

Since we are comparing the means of a continuous variable (crime rate change) across three independent groups (Low, Medium, High gentrification), we use a one-way ANOVA—the standard parametric test for this setting.

Table 4: One-Way ANOVA: Crime Rate Change Across Gentrification Levels

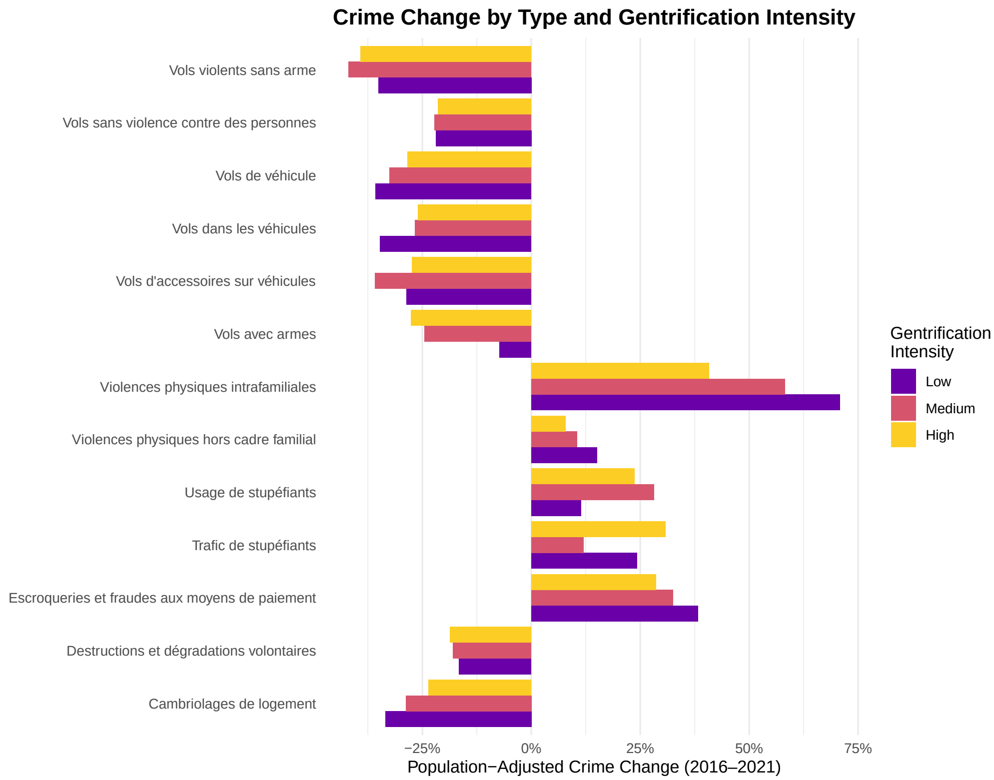
Statistic	Value
F-statistic	0.7580
Degrees of freedom (groups)	2.0000
Degrees of freedom (residuals)	535.0000
P-value	0.4692

The results indicate no statistically significant difference in crime rate changes across gentrification levels ($p > 0.05$). Despite the visual variation observed in the previous figure, we cannot reject the null hypothesis that all three groups experienced similar crime rate changes.

Comparing crime rate change by type among gentrification levels

While total crime rates show only a diminution for all gentrification levels, different crime categories may respond differently to socioeconomic changes. We therefore disaggregate crime into 13 specific types and examine how each evolved across gentrification intensities between 2016 and 2021.

For each city, we compute the population-adjusted change in crime rates for each of the 13 crime categories, thus accounting for both changes in crime counts and changes in population size, providing a cleaner measure of crime-rate evolution.



For each individual crime type, the direction of change (increase or decrease) is consistent across all three gentrification categories. However, the magnitude of change varies considerably depending on the gentrification intensity for certain crime types.

Kruskal-Wallis tests were conducted for each of the 13 crime categories to assess whether population-adjusted crime rate changes differ significantly across gentrification levels.

Only one crime type showed a statistically significant difference: **domestic physical violence** (Violences physiques intrafamiliales), with $X^2 = 19.09$, $p < 0.001$. This indicates that gentrification intensity significantly predicts how domestic violence evolved (positively) between 2016 and 2021.

Crime Type	Chi-squared	P-value	N	Significant
Violences physiques intrafamiliales	19.0937	0.0001	538	Yes
Vols dans les véhicules	5.7361	0.0568	534	No
Escroqueries et fraudes aux moyens de paiement	5.3883	0.0676	538	No
Cambriolages de logement	4.9925	0.0824	533	No
Trafic de stupéfiants	4.7866	0.0913	306	No
Usage de stupéfiants	4.2451	0.1197	524	No

Crime Type	Chi-squared	P-value	N	Significant
Vols d'accessoires sur véhicules	4.1052	0.1284	509	No
Vols de véhicule	3.3616	0.1862	529	No
Vols violents sans arme	2.6530	0.2654	368	No
Vols avec armes	1.7848	0.4097	95	No
Violences physiques hors cadre familial	1.4844	0.4761	538	No
Vols sans violence contre des personnes	1.0988	0.5773	538	No
Destructions et dégradations volontaires	0.6342	0.7282	538	No

Violent vs Non-violent breakdown

Using the violent and non-violent crime categories defined in the descriptive analysis section, we now examine whether gentrification affects these two broad categories differently.

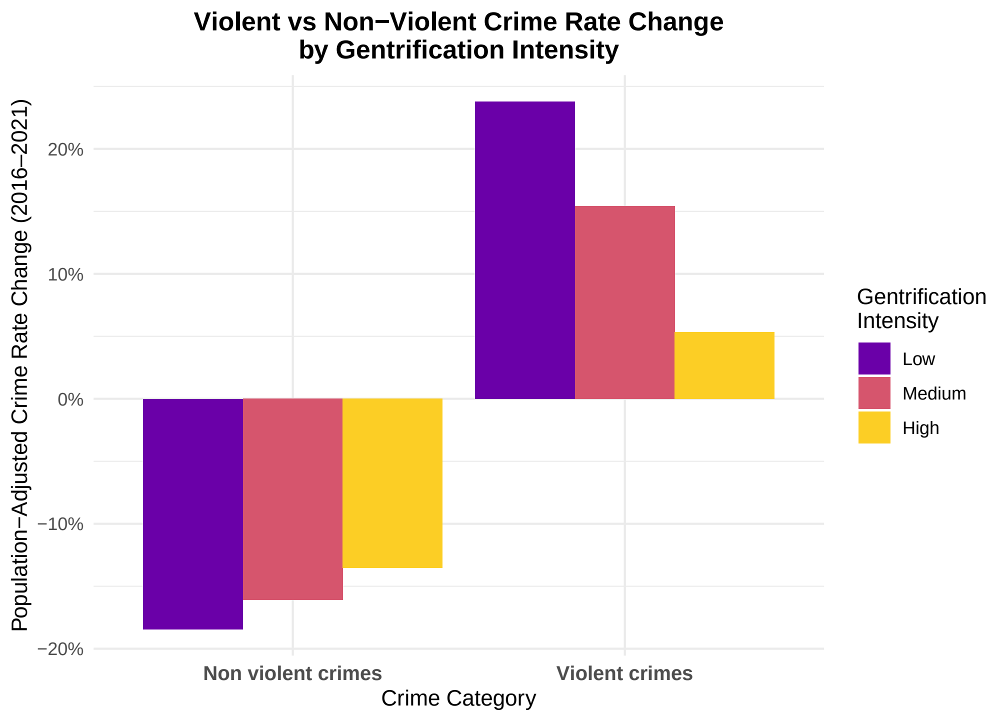


Table 6: Violent vs Non-Violent Crime Rate Change by Gentrification Level

Crime Category	Gentrification Level	Median Change	N Observations
Non violent crimes	Low	-0.184	1120

Crime Category	Gentrification Level	Median Change	N Observations
Non violent crimes	Medium	-0.161	2270
Non violent crimes	High	-0.136	1159
Violent crimes	Low	0.238	363
Violent crimes	Medium	0.154	750
Violent crimes	High	0.053	426

The results reveal a clear gradient pattern: higher gentrification is associated with smaller crime changes in both directions.

Violent crimes increased everywhere, but the increase was much smaller in high-gentrification cities (+5%) compared to low-gentrification cities (+24%).

Non-violent crimes decreased most in low-gentrification cities (-18%), moderately in medium-gentrification cities, and least in high-gentrification cities.

This suggests that gentrification has a stabilizing effect on crime rates, dampening both the rise in violent crime and the decline in non-violent crime. Gentrifying areas experience more moderate changes compared to non-gentrifying areas.

To verify whether these differences are statistically significant, we conduct pairwise Wilcoxon tests with Bonferroni correction. This correction adjusts p-values to account for multiple comparisons, reducing the risk of false positives.:

Table 7: Pairwise Wilcoxon Tests (Bonferroni corrected)

Comparison	Violent crimes (p-value)	Non-violent crimes (p-value)
Low vs Medium	0.2429	0.5520
Low vs High	0.0000	0.0792
Medium vs High	0.0003	0.5866

For violent crimes, high-gentrification cities differ significantly from both low-gentrification ($p < 0.001$) and medium-gentrification cities ($p < 0.001$), confirming that the dampening effect of gentrification on violent crime is statistically robust.

For non-violent crimes, no pairwise comparisons reach statistical significance (all $p > 0.05$), suggesting that the observed differences may be due to chance.

Panel Analysis

The cross-sectional analysis examined the relationship between gentrification and crime changes over the entire 2016-2021 period. However, this approach cannot capture the temporal dynamics of how gentrification affects crime over time. Does gentrification in one year affect crime the following year? Do effects accumulate or reverse over multiple years?

To address these questions, we construct a year-over-year gentrification index that measures annual changes in gentrification intensity for each city. This allows us to test whether lagged gentrification predicts subsequent crime changes using panel data methods.

Year over year gentrification index

Instead of measuring gentrification as the total change from 2016 to 2021, we now compute annual growth rates for each component of the gentrification index. This creates a time-varying measure that captures the intensity of gentrification in each city-year.

The resulting panel contains 2,688 city-year observations across 538 cities from 2017 to 2021. The gentrification index is centered around zero (mean 0) with values ranging from -2.29 to +2.20, indicating substantial variation in annual gentrification intensity across cities and years.

Lagged effects

On total crimes

To test whether gentrification in one year affects crime the following year, we estimate a fixed effects panel model. This approach controls for both time-invariant city characteristics (city fixed effects) and common yearly shocks (year fixed effects).

Table 8: Fixed Effects Panel Model: Lagged Gentrification on Crime Change

Statistic	Value
Coefficient ( )	-0.015
Standard Error	0.007
t-statistic	-2.162
P-value	0.031
R-squared	0.003

The results show a small but statistically significant negative effect (= -0.015, p = 0.03). Cities that experienced more gentrification in one year tended to have slightly smaller increases (or larger decreases) in total crime the following year. However, the effect size is modest: the model explains only 0.3% of the variance in crime changes ($R^2 = 0.003$).

On violent and non violent crimes

Given that our cross-sectional analysis found significant effects specifically for violent crimes, we now examine whether this pattern holds in the panel setting. We test whether gentrification in year t predicts changes in violent and non-violent crime in year t+1.

Table 9: Lagged Gentrification Effect on Crime (1-year lag)

Crime Type	Correlation	P-value	N
Violent crimes	-0.1018	0	2150
Non-violent crimes	-0.1784	0	2150

Violent crimes show a significant negative correlation ($r = -0.102$, $p < 0.001$): cities that experienced more gentrification in one year tended to have smaller increases (or larger decreases) in violent crime the following year.

Non-violent crimes show a stronger and also significant negative correlation ($r = -0.178$, $p < 0.001$): gentrification is associated with subsequent reductions in non-violent crime as well.

Interestingly, the lagged effect is stronger for non-violent crimes than for violent crimes in the panel setting, which contrasts with our cross-sectional findings. This suggests that while gentrification's immediate (same-period) effect may differ by crime type, its delayed effect tends to reduce both categories of crime.

Multiple lags analysis

Table 10: Gentrification Effects on Crime at Different Lags (Correlation, P-value)

Crime Type	1 year	2 years	3 years
Total	-0.182 (< 0.001)	0.341 (< 0.001)	-0.168 (< 0.001)
Violent	-0.102 (< 0.001)	0.143 (< 0.001)	-0.075 (0.0142)
Non-violent	-0.178 (< 0.001)	0.344 (< 0.001)	-0.168 (< 0.001)

This oscillating pattern suggests that gentrification's effect on crime is not permanent. The initial crime reduction is partially reversed in the short-medium term, possibly reflecting displacement effects, adjustment dynamics, or statistical mean reversion. Notably, total and non-violent crime follow nearly identical patterns, while violent crime shows the same directional effects but with smaller magnitudes.

To ensure robustness, we estimate fixed effects panel models controlling for year effects and mean reversion in crime rates. Year fixed effects are particularly important given that 2020 saw an unusual decrease in crime likely due to COVID-19 lockdowns, which could otherwise confound our results. The models include gentrification at 1-year and 2-year lags; we do not extend to 3 lags as our 6-year panel (2016-2021) would leave too few observations for reliable estimation.

Table 11: Fixed Effects Panel Models with Year Controls and Mean Reversion

Crime Type	Variable	Estimate	P-value	Significant
Total	Gentrification (Lag 1)	-0.019	0.022	Yes
Total	Gentrification (Lag 2)	0.000	0.959	No
Total	Mean reversion	-0.468	< 0.001	Yes
Violent	Gentrification (Lag 1)	-0.027	0.02	Yes
Violent	Gentrification (Lag 2)	-0.022	0.054	Borderline
Violent	Mean reversion	-0.532	< 0.001	Yes
Non-violent	Gentrification (Lag 1)	-0.017	0.052	Borderline
Non-violent	Gentrification (Lag 2)	0.006	0.525	No
Non-violent	Mean reversion	-0.468	< 0.001	Yes

The results confirm that gentrification has a modest but significant crime-reducing effect at the 1-year lag, particularly for total crime (= -0.019, p = 0.022) and violent crime (= -0.027, p = 0.020). The effect on non-violent crime is borderline significant (p = 0.052).

Interestingly, only violent crime shows a potential cumulative effect at the 2-year lag (= -0.022, p = 0.054), suggesting that gentrification's impact on interpersonal violence may persist longer than its effect on property crime.

All models show strong mean reversion (-0.50), indicating that crime rates naturally oscillate over time. After controlling for this, gentrification explains a small but statistically significant portion of crime changes.

Robustness checks

Granger Causality

Our analysis assumes that gentrification affects crime, but the reverse could also be true: areas with declining crime might attract more investment and gentrification. To test the direction of causality, we estimate two models for each crime type:

Does lagged gentrification predict crime changes? Does lagged crime change predict gentrification?

Table 12: Granger Causality Tests by Crime Type

Crime Type	Direction	Estimate	P-value
Total	Gentrification → Crime	-0.015	0.031
Total	Crime → Gentrification	0.054	0.523
Violent	Gentrification → Crime	-0.020	0.052
Violent	Crime → Gentrification	0.050	0.387
Non-violent	Gentrification → Crime	-0.015	0.051
Non-violent	Crime → Gentrification	0.034	0.666

The results suggest an asymmetric causal relationship. Gentrification significantly predicts subsequent crime reductions for total crime ($p = 0.031$), with borderline effects for violent ($p = 0.052$) and non-violent crime ($p = 0.051$). Conversely, lagged crime changes do not significantly predict subsequent gentrification (all $p > 0.38$). This asymmetry suggests that the causal direction runs primarily from gentrification to crime—gentrifying areas experience reduced crime growth, but declining crime does not appear to attract gentrification. This strengthens the interpretation that gentrification has a genuine, albeit modest, crime-reducing effect.

Spatial spillover effects

Does gentrification in neighboring cities affect local crime? To test this, we compute the average gentrification index of other cities within the same department (excluding the city itself) and examine whether this "neighbor gentrification" predicts local crime changes. We also control for department-level crime trends to ensure that spillover effects are not simply capturing regional dynamics.

Table 13: Granger Causality Tests by Crime Type

Crime Type	Direction	Estimate	P-value
Total	Gentrification → Crime	-0.015	0.031
Total	Crime → Gentrification	0.036	0.673
Violent	Gentrification → Crime	-0.020	0.052
Violent	Crime → Gentrification	-0.032	0.583
Non-violent	Gentrification → Crime	-0.015	0.051
Non-violent	Crime → Gentrification	0.047	0.561

After controlling for city fixed effects, year fixed effects, and department crime trends, neither own gentrification nor neighbor gentrification significantly predicts crime changes. This suggests that there is no evidence of spatial spillovers or crime displacement to neighboring areas once we account for common temporal shocks at the city level. This result however has huge limitations and shouldn't be interpreted as "gentrification doesn't have any spatial spillover effect" (see conclusion).

Conclusion

This study examined the relationship between gentrification and crime rate evolution in 897 French cities from 2016 to 2021, using a weighted gentrification index based on real estate prices, median income, income distribution (D1 and D9), and poverty rates.

Cross-sectional findings

The cross-sectional analysis revealed no significant linear correlation between gentrification intensity and overall crime rate changes ($r = 0.052$, $p = 0.227$). Similarly, ANOVA tests found no statistically significant differences in total crime rate changes across low, medium, and high gentrification categories. However, disaggregating by crime type uncovered important heterogeneity. Among 13 crime categories tested, only domestic physical violence showed significant variation across gentrification levels ($^2 = 19.09$, $p < 0.001$).

When aggregating into violent versus non-violent categories, a clear gradient emerged: violent crimes increased everywhere during this period, but substantially less in highly gentrified cities (+5%) compared to low-gentrification cities (+24%), a difference that is statistically significant ($p < 0.001$). Non-violent crime differences, while visually apparent, did not reach statistical significance.

Panel analysis findings

The year-over-year panel analysis strengthened these findings by exploiting temporal variation. Fixed effects models controlling for city-specific characteristics and common yearly shocks revealed a modest but statistically significant crime-reducing effect of gentrification at the 1-year lag for total crime ($= -0.019$, $p = 0.022$) and violent crime ($= -0.027$, $p = 0.020$). The effect on non-violent crime was borderline significant ($p = 0.052$). Notably, only violent crime showed a potential cumulative effect persisting to the 2-year lag ($= -0.022$, $p = 0.054$), suggesting that gentrification's impact on interpersonal violence may be more durable than its effect on property crime.

The multiple-lag correlation analysis revealed an oscillating pattern: gentrification initially reduces crime rates (lag 1), but this effect partially reverses at the 2-year lag before returning to a negative relationship at 3 years. This non-monotonic dynamic suggests that gentrification's impact on crime involves complex adjustment processes—possibly reflecting displacement effects, changes in neighborhood social dynamics, or statistical mean reversion—rather than a simple permanent reduction.

Causality and directionality

Granger causality tests provided evidence supporting a primarily unidirectional relationship. Lagged gentrification significantly predicted subsequent crime reductions for total crime ($p = 0.031$), with borderline effects for violent ($p = 0.052$) and non-violent crime ($p = 0.051$). Crucially, the reverse relationship was not significant: lagged crime changes did not predict subsequent gentrification (all $p > 0.38$). This asymmetry strengthens the interpretation that gentrification exerts a genuine, albeit modest, crime-reducing effect rather than simply being attracted to areas where crime is already declining.

Spatial spillovers

After controlling for city fixed effects, year fixed effects, and department-level crime trends, we found no evidence of crime displacement to neighboring cities within the same department. However, this null result should be interpreted with considerable caution. Our spatial definition—averaging gentrification across all other cities in the same department—is geographically coarse and may miss localized spillovers between immediately adjacent municipalities. Additionally, crime displacement could occur at finer spatial scales (neighborhoods within cities) or to areas outside department boundaries, neither of which our data can capture.

Limitations

Several constraints temper the interpretation of these findings. First, the 6-year panel (2016–2021) limits detection of longer-term effects and includes the anomalous COVID-19 period, during which crime patterns were distorted by lockdowns and mobility restrictions. Second, data coverage excludes important regions (Alsace, Moselle) due to different legal frameworks, and lacks arrondissement-level detail for France's largest cities (Paris, Marseille, Lyon), precisely where gentrification dynamics may be most pronounced.

Third, our analysis relies on police-recorded crime statistics, which reflect not only actual criminal activity but also reporting behavior, victim willingness to file complaints, and police enforcement priorities. This concern is particularly relevant for domestic violence—the only crime type showing significant cross-sectional differences—where reporting rates have increased substantially due to social movements and policy initiatives during our study period.

Fourth, the gentrification index, while theoretically grounded, could be enriched with additional indicators such as educational attainment, occupational structure, housing tenure changes, or commercial establishment turnover.

Fifth, our analysis operates at the city level, which may mask substantial intra-urban heterogeneity. Gentrification typically occurs at the neighborhood level, with some districts experiencing intense transformation while others remain unchanged. City-wide averages dilute these localized dynamics and may obscure stronger relationships that exist at finer spatial scales. The effects we identify—or fail to identify—at the municipal level may not generalize to neighborhood-level analyses, where gentrification's impact on crime could be more pronounced, more localized, or even reversed due to micro-level displacement within city boundaries.

Finally, despite our panel methods and robustness checks, unobserved confounders correlated with both gentrification and crime trends cannot be entirely ruled out.

Policy implications

Despite these limitations, our findings carry implications for urban policy. Gentrification appears to have a stabilizing effect on crime rates, dampening both the increase in violent crime and—to a lesser extent—the decrease in non-violent crime observed nationally during this period. The effect is most robust for violent crime, suggesting that the social transformations accompanying gentrification (increased surveillance, changed street-level dynamics, shifts in city's composition) may particularly affect interpersonal violence.

However, policymakers should not overstate these findings. The modest effect sizes ($R^2 < 1\%$ in panel models) indicate that gentrification explains only a small fraction of crime variation. city gentrification alone will not substantially reduce crime, and the social costs of gentrification—including displacement of vulnerable populations, erosion of community networks, and increased housing insecurity—must be weighed against any crime-reduction benefits. Future research should examine whether displaced populations experience increased crime exposure in their new locations, a question our department-level spillover analysis cannot adequately address.

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